

Abstract

The latency of writing to PCM is higher than reading and is sensitive to the data to be written. This paper finds that overwriting unknown memory content can incur significantly higher latency and energy compared to overwriting known all-zeros or all-ones content. Based on this, this paper proposes data content aware PCM writes (DATACON), which can reduce the latency and energy of PCM writes by redirecting these requests to overwrite memory locations containing all-zeros or all-ones.

Introduction

DRAM is increasingly unable to meet the rising computational demands of future systems due to significant challenges in technology scaling. As a result, numerous emerging memory technologies, such as ReRAM, CBRAM, FeRAM, and PCM, are being actively explored as potential alternatives. Among them, PCM is a relatively mature technology; however, it still faces challenges, particularly regarding limited write endurance and higher access latency compared to DRAM. One key characteristic of PCM, as mentioned in this paper, is the asymmetry between its read and write latencies.

This paper notes that modern computing systems are adopting hybrid memory architectures that combine DRAM and PCM to leverage the strengths of both technologies. However, these approaches do not fundamentally reduce PCM write latency, leaving PCM latency as an unavoidable performance bottleneck.

To reduce the latency and energy of PCM writes, this paper leverages the following three key observations:

1. The energy consumed by a PCM write depends on both the write data and the original content being overwritten. When the proportion of SET bits in the write data is below 60%, overwriting an all-0s block is more energy-efficient than overwriting an all-1s block.
2. The number of SET bits in the write data varies based on workload. To address this, this paper proposes **Data Content-Aware PCM Writes (DATACON)**, which estimates the potential energy savings of overwriting either all-0s or all-1s content. DATACON then redirects write requests to memory locations with optimal pre-existing content for overwriting. Notably, DATACON processes each write request using only SET or RESET operations, enhancing system performance.
3. Many workloads exhibit a form of locality where rows within the same memory partition are frequently accessed. This paper refers to this as Partition Level Spatial Locality (PLSL) and leverages it to reduce the overhead of the address

translation table by caching only the address translations of frequently accessed PCM partitions.

Background

In this section, the paper describes some knowledge related to PCM Operation and PCM Timings, such as **ACTIVATE**, **READ/WRITE** and **PRECHARGE**. Each command causes a different but fixed latency which is independent of the data content.

With DATACON, latency varies depending on whether the data content is known. If the data content is known, latency is reduced; otherwise, if the data content is unknown, latency remains unchanged. This is because, when the data content is known, we can just use SET or RESET to write data, focusing only on the bits that need to be overwritten.

Motivation

The paper presents examples of potential improvements. When writing data to a location with unknown content, servicing the write request requires 4 steps:

1. Compare the write data with the existing content to identify the PCM cells that need to be set.
2. Set the selected PCM cells (long latency).
3. Compare the write data with the existing content again to identify the PCM cells that need to be reset.
4. Reset the selected PCM cells (short latency).

However, if the overwritten content is already known (e.g., all-1s or all-0s), only half of these steps are required, resulting in enhanced performance and reduced energy consumption.

There is a trade-off between using SET to overwrite all-0s and using RESET to overwrite all-1s. RESET operations are good for performance, while SET operations are good for energy. The exact energy and performance are related to the number of bits that are to be SET or RESET when overwritten content. **These motivations present opportunities for improvement, and DATACON leverages these ideas to enhance the performance of PCM.**

When the number of SET bits in the write data is more than 60%, DATACON will first select the all-1s location as the overwritten content. This is both more energy efficient and higher performance. Otherwise DATACON select the all-0s locations as the overwritten content to achieve lower latency than overwriting unknown content.

When the number of SET bits in the write data is less than 60%, DATACON will first select the all-0s location to achieve more energy efficient. Otherwise DATACON select the all-1s locations as the overwritten content to achieve lower latency than overwriting unknown content. If neither all-0s nor all-1s locations are available in PCM, DATACON will choose the unknown content location to service the write request.

Evaluation

This paper employs a DRAM-PCM hybrid memory simulator to evaluate DATACON using a variety of workloads: 8 from SPEC CPU2017, 2 from NAS Parallel, and 10 TensorFlow machine learning tasks, each executing 10 billion instructions. The evaluated techniques include Baseline, PreSET, Flip-N-Write, and DATACON.

By methodically re-initializing unused memory locations and utilizing only SET or RESET operations to handle write requests, DATACON achieves the highest performance improvement among the evaluated techniques. Specifically, its execution time is 47% lower than Flip-N-Write, 27% lower than PreSET, and 40% lower than the Baseline.

Furthermore, DATACON demonstrates reduced access latency, being 43% lower than Baseline, 50% lower than Flip-N-Write, and 31% lower than PreSET.

In terms of energy consumption, DATACON takes advantage of the energy-latency trade-offs of SET and RESET operations, resulting in an average energy consumption that is 27% lower than Baseline, 26% lower than Flip-N-Write, and 43% lower than PreSET.

My thoughts

The experiments conducted in this paper are quite comprehensive, covering various assessments of performance, latency, energy consumption, re-initialization overhead, LUT sizing, and PCM lifetime.

This paper also discusses other scenarios with DATACON, such as encryption. While this topic is not the focus of the paper, it enriches the content and gives me a better understanding of the potential future applications of DATACON.